

**MINISTRY OF EDUCATION AND TRAINING
NATIONAL ECONOMICS UNIVERSITY**



**GROUP ASSIGNMENT
SUBJECT: ECONOMETRICS 1**

TOPIC: IDENTIFYING DETERMINANTS OF NET MIGRATION
IN VIETNAMESE SOUTHERN PROVINCES FROM 2015 TO 2024

Class :	Actuary 66B	
Instructor :	Dr. Bui Duong Hai	
Group 6:	1. Than Phuc Khang	11245768
	2. Duong Ngoc Hai	11245755
	3. Nguyen Dinh Nguyen	11245790
	4. Nguyen Duc Hai Nam	11245788

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SELF EVALUATION TABLE

ID	Full name	Contribution	(%)	Sig.
11245768	Than Phuc Khang	- Group Leader: Planning, scheduling, and task management. - Brainstorming research ideas and conceptualization. - Data collection and Reference sourcing. - Executing model fitting (R-studio) and model selection. - Authoring Sections 5, 6, 7 ,8 and 9(Results, Analysis, Conclusion & Limitation, Reference, Appendix).	40%	
11245755	Duong Ngoc Hai	- Data research and Reference sourcing. - Authoring Section 1 (Introduction) and Section 2 (Research Questions).	20%	
11245790	Nguyen Dinh Nguyen	- Idea brainstorming. - Data research and Reference sourcing. - Authoring Section 3 (Theoretical Framework and Literature Review).	20%	
11245788	Nguyen Duc Hai Nam	- Data research and Reference sourcing. - Authoring Section 4 (Data and Methodology) under leader's guidance.	20%	
TOTAL			100%	

SUPERVISOR EVALUATION TABLE

	A	B	C	D	E
Purposes and questions					
Literature reviews					
Data and Model					
Estimated result					
Finding/comment					
Formating					

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1. INTRODUCTION

Internal migration has become a key driver of Vietnam's economic transformation, particularly through labor flows from the Mekong River Delta to the Southeast region. This study focuses on six representative provinces (Bac Lieu, Soc Trang, An Giang, Kien Giang, Ho Chi Minh City and Ba Ria - Vung Tau) to better capture regional differences in migration patterns.

Between 2015 and 2025, migration in Vietnam shows a clear directional trend toward the Southeast, which consistently records the highest net in-migration rate. In contrast, provinces in the Mekong River Delta experience persistent net out-migration, reflecting ongoing regional imbalances. These migration patterns are largely driven by economic disparities. The Mekong River Delta is characterized by limited non-agricultural employment, lower income levels, and environmental pressures such as climate change. Meanwhile, Ho Chi Minh City and Ba Ria–Vung Tau offer better job opportunities, higher wages, and more developed infrastructure. As a result, migration flows from provinces such as Soc Trang or Bac Lieu to Ho Chi Minh City are significantly larger than reverse flows.

From a theoretical perspective, this reflects the push–pull mechanism, where migration decisions depend on relative economic conditions between origin and destination. For example, many young workers from the Mekong River Delta migrate to urban centers to work in industrial and service sectors. Focusing on provincial-level data allows for a more precise econometric analysis and avoids aggregation bias associated with national-level data. This study contributes to the literature by providing a province-level analysis of migration between key sending and receiving regions.

Based on this context, this study aims to empirically examine how unemployment rate (Unem), urbanization level (Urb), and population density (Pd) influence migration decisions between the Mekong River Delta and the Southeast region.

2. RESEARCH QUESTIONS

Our modeling journey was steered by two fundamental questions:

- (1) How do population density and urbanization levels influence the net migration patterns across Southern Vietnam's provinces?
- (2) How does each identified factor impact the Net Migration of the studied provinces?

3. THEORETICAL FRAMEWORK AND LITERATURE REVIEW

3.1. Theoretical Framework

This study is mainly based on Lee's Push–Pull Theory of Migration. Lee (1966) explains that migration decisions are shaped by conditions at the place of origin, conditions at the place of destination, intervening obstacles, and personal factors. In this view, migration occurs when the expected benefits of moving are greater than the costs of staying.

This framework is useful for our study because it provides a simple basis for linking migration to urban development, population pressure, migration persistence, and regional disadvantages. Provinces with stronger urban and economic conditions may attract migrants, while provinces facing structural pressure may lose population.

However, Lee's framework also has clear limits in the current context. It was developed before digital job search, online information, and stronger migration networks reduced information barriers across regions. It also does not directly capture newer structural forces such as environmental stress, climate vulnerability, and long-term livelihood change, which are increasingly important in Vietnam, especially in the Mekong Delta. Therefore, Lee's theory is useful as a starting point, but it is not enough on its own to explain current migration patterns.

3.2. Literature Review and Research Gap

Previous studies show that internal migration in Vietnam is mainly shaped by urban development, population concentration, migration persistence, and regional differences. However, these factors are rarely examined together in one province-level net migration model. This is the main gap that our study addresses.

First, earlier studies consistently suggest that urban development is a major pull factor. Migrants tend to move toward provinces that are more urbanized, industrialized, and economically dynamic (Dang et al., 1997; UNFPA, 2020). This gives a clear basis for including urban development in our model. At the same time, earlier studies do not fully test whether urban development still matters once migration persistence and region-specific pressure are considered together.

Second, previous studies suggest that population concentration matters, but the effect of density is not straightforward. Research in Vietnam supports the role of economic hubs and urban concentration more clearly than population density as a separate standard regressor (Huynh Truong & Nonneman, 2012; Nguyen & Phung, 2020). This suggests that density alone may not be enough. Its effect may depend on whether concentration is supported by urban development or weakened by congestion.

Third, earlier studies also indicate that migration has a persistent character. Once migration flows begin, they may continue because earlier migrants create information channels and social networks for later movers (Nguyen-Hoang & McPeak, 2010; Huynh Truong & Nonneman, 2012). However, this persistence is not always incorporated directly into empirical migration models.

Fourth, previous studies show that regional differences matter beyond standard economic variables, especially in the Mekong Delta. Recent evidence suggests that this region faces overlapping pressures from agricultural transition, livelihood vulnerability, environmental stress, and strong labor outflows toward the Southeast (World Bank, 2025; Tran et al., 2023). This means that migration patterns in the Mekong Delta may not be fully explained by urban development or population concentration alone. However, earlier studies do not always include this regional pressure directly in the model.

Overall, earlier studies help explain why migration in Vietnam is linked to urban development, concentration effects, persistence, and regional inequality. However, most existing studies treat these factors independently, overlooking their potential joint effects or regional variations. Our study contributes by bringing these factors into a single framework and testing how they jointly shape provincial net migration.

4. DATA AND METHODOLOGY

4.1. Data Description and Variable Rationalization

- Population Density (Pd)

This is included based on the resource scarcity hypothesis. High population density exerts substantial pressure on land availability, infrastructure, and the living environment. In the context of the Push-Pull theory, Pd acts as a critical "push" factor, particularly in agrarian-heavy regions where the fragmentation of productive assets (such as land) limits economic viability for the local population.

- Urbanization Ratio (Urb)

This variable represents the structural transformation of a region, signaling superior infrastructure, enhanced public services (such as healthcare and education), and modern amenities. High urbanization rates generate a magnetic "pull" effect, as individuals anticipate a higher quality of life and better access to intellectual and social capital in urban centers compared to rural origins.

- The logarithm of the urban ratio 'Log(Urb)'

This variable is included to capture the effect of urban development in a more flexible form. Provinces with a higher urban ratio usually offer broader job opportunities, stronger market links, and better access to services, which may increase their attractiveness to migrants. The logarithmic form is used because the effect of urbanization may not increase at the same rate across all provinces.

- The interaction term (PdUR)

It is measured as population density multiplied by the urban ratio, and is included to examine whether the migration effect of population density depends on the level of urbanization. A high-density province may attract migrants when urbanization is high, because density may reflect stronger economic concentration, larger labor markets, and better access to services. However, high density may also create congestion, housing pressure, and overloaded infrastructure. The interaction term therefore captures the joint effect of density and urbanization rather than treating them as fully separate forces.

- The interaction term (UNUR)

It is measured as the unemployment rate multiplied by the urban ratio, and is included to examine whether the migration effect of unemployment depends on the level of urbanization. A high-unemployment province may trigger stronger out-migration when urbanization is high, as urban dwellers often have better access to job information and lower costs to move between economic

centers. In contrast, in less urbanized regions, people may find it harder to leave despite being unemployed due to limited mobility. The interaction term therefore captures the joint effect of unemployment and urbanization rather than treating them as fully separate forces.

- OMLAG

It is included to capture persistence in migration behavior, the dynamic nature of migration. If a province has already experienced strong out-migration, that pattern may continue because earlier migration flows lower information costs and strengthen social networks for later movers. In this sense, the variable reflects migration momentum.

- DumDBSCL

It is included as a regional control for the Mekong Delta. It captures structural pressures that may not be fully explained by the other regressors, such as agricultural transition, environmental stress, and long-term labor outflows toward the Southeast. This is important because migration in the Mekong Delta may reflect not only economic opportunity, but also broader regional vulnerability.

- The Ho Chi Minh City Metropolitan Dummy (HCMC)

Its "pull" force is disproportionately stronger than any other urban area due to the extreme concentration of capital, foreign investment, and industrial zones. Including this dummy helps prevent the model from being skewed by the city's massive migration figures. It allows the model to distinguish between general urbanization effects and the unique "super-magnet" effect of Vietnam's largest commercial hub.

The data of NM, Omlag, Pd, Unem, Urb are collected from the Statistical Yearbook of General Statistic Office of Viet Nam (GSO) from 2015 to 2024. Table I presents a statistical summary of these variables

Table I. Statistical table

Variables	Definition	Formula	Measurement	Expected sign
Pd	Population density		ratio	+
Urb	Urban ratio		ratio	+
Log(Urb)	Log of Urb	Log(Urb)		-
PdUr	Interaction between Pd and Ur	$Pd*Ur$		+
UnUr	Interaction between Un and Ur	$Un*Ur$		+
Omlag	Out migration from the second year			+
DumHCM	HCM or not			+
DumDBSCL	DBSCL or not			-

4.2. Methodology:

This research is conducted to study the determinants of out-migration across provinces in Vietnam, including key variables: population density (Pd), urban ratio (Urb), the lagged out-migration (Omlag), and the interaction terms (PdUr and UnUr). To capture regional characteristics, dummy variables for Ho Chi Minh City (DumHCM) and the Mekong Delta (DumDBSCL) are also included.

The null and alternative hypotheses are:

$$\begin{cases} H_0: \beta_i = 0 \text{ (Independent variables have no explanatory power on Net Migration)} \\ H_1: \beta_i \neq 0 \text{ (Independent variables have an explanatory power on Net Migration)} \end{cases}$$

Our main regression function is:

$$\text{NetMigration} = f(\text{Pd}, \text{Urb}, \text{Log}(\text{Urb}), \text{PdUr}, \text{UnUr}, \text{Omlag}, \text{Dum HCM}, \text{DumDBSCL})$$

The results of the coefficients β_i will indicate the relationship between the independent and dependent variables, being either negative or positive. P-values of the coefficients will be compared with significance levels of $\alpha = 10\%$, 5% , and 1% and then rated “*”, “**”, and “***”, respectively.

We used R-studio as the tool for model fitting and an environment for all of our statistical computing and graphics. Variables were selected based on economic theory and statistical significance, and we used the OLS method for estimation of coefficients and standard errors. Then each model underwent diagnostic tests, namely: Overall significance test (F-test), Ramsey’s RESET test, Multicollinearity test (VIF), and Heteroscedasticity test (Breusch-Pagan test). Finally, we used a comparison table to decide which of the models was the most suitable for explaining migration patterns.

5. MODEL SELECTION

5.1. Model Comparison Table

Variable	Model 1	Model 2	Model 3	Model 4
Intercept	-5.9023***	-1.7632	-9.6237***	14.0526
Pd	-	-	11.7036***	3.5654***
Urb	-	-0.2667***	-0.0443	-
Log(Urb)	-	-	-	-4.1847*
PdUR (Interaction)	0.0387***	0.2784***	-	-
UNUR	0.0158*	-	-	-
Omlag	-0.3917***	-0.2594**	-0.2663**	-0.2519**

DumHCM	-	-61.5022***	-26.5208***	-
DumDBSCL	-	-	-	-7.7320***
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R-squared	0.6668	0.712	0.712	0.7011
Adjusted R-squared	0.6536	0.6966	0.6966	0.6851
F-statistic	50.69	46.35	46.35	43.97
p-value (F-Test)	< 2.2e-16	< 2.2e-16	< 2.2e-16	< 2.2e-16
Residual Std. Error (sdv)	4.652	4.354	4.354	4.435
RMSE	4.5342	4.2154	4.2154	4.294594
MAE	3.6403	3.1827	3.2233	3.211971
MAPE	106.7331	76.6307	72.32331	75.04908
RESET test (p-value)	0.5543	0.4405	0.4405	0.1011
BP test (p-value)	0.08241	0.1172	0.1172	0.1227

Variable	Model 1	Model 2	Model 3	Model 4
VIF: Urb	-	15.989878	5.506879	
VIF: Pd	-	-	48.265847	2.782327
VIF: Log(Urb)	-	-	-	5.504309
VIF: PdUR	2.3491	196.683702	-	-
VIF: Omlag	1.1628	1.3131	1.300283	1.397100
VIF: UNUR	2.2303	-	-	-
VIF: DumHCM	-	118.966573	32.051667	-
VIF: DumDBSCL	-	-	-	6.153961

5.2. Model Selection Process

The development of our final model was a step-by-step process of balancing statistical power with model reliability. We began with Model [1], which served as our baseline. At first glance, this model seemed quite promising because all included variables were statistically significant (p -value < 0.1) and the R^2 was a solid 0.6668. However, as we looked deeper, we realized this model was too simple. It only captured basic linear relationships and failed to account for regional differences or non-linear trends. Most importantly, the error rate was extremely high, with a MAPE of 106.73%, meaning our predictions were often off by more than 100%. While Model [1] was useful for a general comparison of how variables affect the dependent factor, it wasn't accurate enough for a high-quality report.

To improve the accuracy, we moved to Model [2] and Model [3], where we introduced dummy variables for specific regions like Ho Chi Minh City (DumHCM) and interaction terms like Population Density multiplied by Urban ratio (PdUR). While these additions successfully pushed our R^2 up to 0.712 and reduced the overall error, they created a massive technical problem: Multicollinearity. In Model 2, the Variance Inflation Factor (VIF) for some variables exploded to over 196, and Model 3 still showed VIFs above 48. In econometrics, such high VIF values mean our independent variables were too closely related to each other, making our coefficient estimates unstable and unreliable. Even though the "numbers" looked better on the surface, the internal structure of these models was broken.

Finally, we arrived at Model [4], which we believe is the most robust and "optimal" choice for our analysis. We made a strategic decision to use the natural logarithm of Urbanization (Log Urb) instead of its linear form, which helped stabilize the data and better reflect real-world marginal effects. The breakthrough in Model [4] was the successful control of multicollinearity; we managed to bring all VIF values down to a safe range (below 7), ensuring our results are statistically trustworthy. Furthermore, Model [4] passed the RESET test ($p = 0.1011$) for functional form and the BP test ($p = 0.1227$) for heteroscedasticity. By choosing Model [4], we sacrificed a tiny bit of R-squared compared to Model 2, but we gained a model that is statistically sound, theoretically logical, and much more reliable for drawing final conclusions.

We can now write the regression model:

$$NM = \beta_0 + \beta_1 * Pd + \beta_2 * \log(Urb) + \beta_3 * Omlag + \beta_4 * DumDBSCL$$

6. ANALYSIS

Following our selection journey, this section provides a detailed technical evaluation of Model 4, which has been identified as our most reliable model.

6.1. Model Specification and Estimation

- The relationship between our independent variables and the dependent variable is represented by the following econometric model:

$$NM = \beta_0 + \beta_1 * \log(Urb) + \beta_2 * Pd + \beta_3 * Omlag + \beta_4 * DumDBSCL + \varepsilon$$

- The estimated regression equation for Model 4 is specified as follows (standard errors are provided in parentheses below each coefficient):

$$\widehat{NM} = 14.053 - 4.185 * \log(Urb) + 3.565 * Pd - 0.252 * Omlag - 7.732 * DumDBSCL$$

- Estimated Coefficients & Significance:

	Intercept	Log(Urb)	Pd	Omlag	DumDBSCL
Se	10.0330	2.4843	0.6474	0.1102	2.8410
P-value	0.16544	0.09626*	4.88e-07 ***	0.02509 **	0.00807 ***

- Population Density (Pd): Highly significant at the 0.1% level ($p < 0.001$), showing a strong positive impact on NM.
- Omlag & DumDBSCL: Both are statistically significant at the 5% and 1% levels, respectively, indicating their importance in explaining the variance of NM.
- Log(Urb): Significant at the 10% level ($p = 0.096$), suggesting a meaningful non-linear relationship between urbanization and the dependent variable.
- While Omlag is treated as an independent variable to capture the persistence of migration trends, we recognize that it may introduce a degree of autoregressive bias. In strictly theoretical terms, this can lead to endogeneity issues where the regressor is correlated with the error term. However, for the purpose of this study, \$Omlag\$ serves as a crucial indicator of Path Dependency, showing that high out-migration in the past creates a momentum that is difficult to reverse. Future research could employ Dynamic Panel Data models (such as GMM) to further refine these estimates.

6.2. Overall Model Significance and Fit

- F-test for Overall Significance:

Hypotheses:

$$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0 \text{ (All slope coefficients are zero).}$$

$$H_1: \exists \beta_j \neq 0 \text{ (The model is significant).}$$

Results: $F_{stat} = 43.97$, $p - value < 2.2e-16$

$$F_{crit} = F_{(80-4-1;4)0,05} = 2.493696$$

The F-statistic is 43.97, which is significantly higher than the critical value F_{crit} at the 5% significance level (with $df_1 = 4$; $df_2 = 80 - 4 - 1 = 75$). Consistently, the p-value is essentially zero ($< 2.2e-16$). Also the p-value is much smaller than our significance level ($\alpha = 0.05$), we reject the null hypothesis H_0 : This confirms that the model is globally significant, meaning the independent variables combined have a valid explanatory power.

- Goodness of Fit:

Our Adjusted R-squared of 0.6851 is a solid result for a social science study. Unlike models with extremely high R-squared values (above 0.95) which might be overfitted to a specific sample, our model stays realistic. It explains nearly 69% of migration movements while remaining statistically safe through VIF and RESET tests. This ensures that our conclusions about the Southern provinces are trustworthy and reflect real-world trends

6.3. Accuracy and Prediction Error

To evaluate the predictive performance and residual magnitude, we calculated the following metrics:

- RMSE (Root Mean Square Error): 4.2946 — This measures the average deviation of the estimates from the actual values in the same units as the dependent variable.
- MAE (Mean Absolute Error): 3.2120 — On an absolute scale, the model's predictions deviate by only about 3.2 units from the actual net migration values. Given the scale of population movement in Southern Vietnam, this represents a highly reliable fit.
- Regarding the MAPE of 75.049%, while numerically high, it does not signify a failure of the model. This is primarily due to the mathematical sensitivity of MAPE when the dependent variable (Net Migration) approaches zero. In provinces where net migration is near-neutral, even a minor absolute residual results in a disproportionately large percentage deviation. Furthermore, the high MAPE reflects the structural polarization of our dataset, which includes both massive industrial hubs like Ho Chi Minh City and agricultural provinces with heavy out-migration. Capturing such extreme variations in a single linear framework is inherently challenging. However, the consistency between our low MAE and the statistically significant F-test confirms that the model is robust and provides a valid foundation for regional analysis.

6.4. Diagnostic Tests (Post-Estimation)

To ensure the validity of our Model assumptions, we conducted several diagnostic tests:

- Multicollinearity Test (Variance Inflation Factor - VIF):
 - Logic: This test evaluates whether the independent variables are highly correlated with each other, which could undermine the statistical significance of individual predictors.
 - Hypothesis pair/ Decision Rule:

H_0 : There is no serious multicollinearity in the model ($VIF < 10$).

H_1 : There is a serious multicollinearity problem ($VIF \geq 10$).

(Note: Although VIF is a diagnostic metric rather than a formal hypothesis test, the threshold of 10 is the standard academic rule for rejection.)

- Results:

VIF: Log(Urb) : 5.50

VIF: Pd : 2.78

VIF: Omlag : 1.40

VIF: DumDBSCL : 6.15

- Conclusion: Since all VIF values are well below the threshold of 10, we conclude that multicollinearity is not a concern in Model 4.
- Ramsey's RESET Test (Functional Form):
- Mathematical Logic:

We test the model by adding the powers of the fitted values ($\widehat{NM}^2, \widehat{NM}^3, \dots$) as auxiliary independent variables:

$$NM = (\beta_0 + \beta_1 * \log(\text{Urb}) + \beta_2 * \text{Pd} + \beta_3 * \text{Omlag} + \beta_4 * \text{DumDBSCL}) + (\alpha_1 * \widehat{NM}^2 + \alpha_2 * \widehat{NM}^3 + \dots) + \varepsilon$$

- Hypothesis pair:
 - $H_0: \alpha_1 = \alpha_2 = \dots = 0$ (The functional form is correctly specified)
 - $H_1: \exists \alpha_j \neq 0$ (At least one non-linear terms is significant, meaning the model is incorrect)
- Results: $F_{stat} = 0.90137, p - value = 0.1011, F_{crit} = 3.12964$
- Conclusion:
 - Since the $p\text{-value} > 0.05, F_{stat} < F_{crit}$, we fail to reject H_0 , there is not enough evidence to say that the functional form is incorrect. This indicates that our model is correctly specified.
- Breusch-Pagan (BP) Test (Heteroscedasticity):
- Hypotheses:
 - H_0 : Homoscedasticity is present (The variance of the error term is constant).
 - H_1 : Heteroscedasticity is present (The variance of the error term is not constant).
- Results: $BP = 7.2615, p - value = 0.1227$
- Conclusion:
 - Since the $p\text{-value} : (0.1227) > 0.05$ (and also > 0.10), we fail to reject H_0 . This confirms that the variance of the residuals is constant, satisfying the OLS assumption of homoscedasticity.

6.5. Research questions and discussion

(1) How do population density and urbanization levels influence the net migration patterns across Southern Vietnam's provinces?

- According to our best result - Model [4], the primary economic drivers of Net Migration (NM) are Population Density (Pd) and Urbanization (Log Urb). Statistical results indicate that both variables have a significant impact on migration decisions. Specifically:
- Population Density acts as a powerful "pull" factor; as density increases, the availability of Infrastructure and job markets naturally attracts a larger workforce.
- Urbanization, however, shows a non-linear relationship with migration. By using the natural logarithm of urbanization, our model reveals that while urban growth initially draws people in, it eventually reaches a "saturation point" in major hubs like Ho Chi Minh City. At this stage, high living costs and congestion begin to slow down the net inflow of migrants.

(2) How does each identified factor impact the Net Migration of the studied provinces?

Looking at the estimated equation of Model [4], we can interpret the results as follows:

- Estimated $\beta_0 = 14.0526$ (Intercept): It represents the theoretical value of Net Migration when all other independent variables (Density =0, log(Urban) =0, etc.) are equal to zero.
- Estimated $\beta_1 = -4.1847$ (Log of Urban ratio): A negative coefficient here means that once a province becomes "too urban," it actually starts to attract fewer net migrants. Although the coefficient for Log(Urb) is statistically significant only at the 10% level, it provides a critical insight into the non-linear nature of urbanization. This suggests that the "pull factor" of cities is not infinite. In hyper-urbanized centers like Ho Chi Minh City, the marginal benefits of urbanization are increasingly offset by negative externalities such as traffic congestion and high cost of living, indicating a potential "saturation point" for net migration.
- Think of Ho Chi Minh City: as urbanization peaks, the city becomes crowded. High living costs, traffic congestion, and a shortage of affordable housing act as "push factors." Instead of everyone piling into the city center, people start moving to surrounding satellite areas where life is cheaper. So, for every 1% increase in the urbanization rate, we see a slight decrease in the net migration flow (about 0.04 units), because the city is essentially "full."
- Estimated $\beta_2 = 3.5654$ (Population Density): This shows a strong positive relationship. Specifically, if a province's density increases by 1,000 people/km², the Net Migration is expected to rise by 3.56%. High-density areas often represent established economic hubs with superior infrastructure and vibrant job markets, which continue to act as a magnet for the regional workforce.
- Estimated $\beta_3 = -0.2519$ (Lagged Out Migration - Omlag): This coefficient indicates a persistent trend in population loss. If the Out-migration rate in the previous period was 1% higher, the current Net Migration tends to decrease by 0.25%. This reflects a 'momentum effect' in migration; once a province begins to lose a significant portion of its workforce (due to local economic decline or environmental factors), it is difficult to reverse that trend in the short term, leading to a continued deficit in the net population flow.
- Estimated $\beta_4 = -7.7320$ (Mekong Delta Dummy - DumDBSCL)

This variable highlights a significant regional disparity. Holding other factors constant, provinces in the Mekong Delta (such as An Giang, Kien Giang, and Dong Thap,...) have a Net Migration level that is 7.73 units lower than other non-Mekong Delta provinces (Ho Chi Minh, Vung Tau). This statistically confirms the "out-migration" trend in the Delta region, where people tend to move toward industrial centers like Ho Chi Minh City or Vung Tau in search of better industrial opportunities.

6.6. Interpretation of Findings

In this section, we provide a deeper interpretation of our econometric results by connecting the statistical outputs with the socio-economic realities of the southern provinces of Vietnam.

- Regional Specificity and Structural "Push" Factors:

The coefficient for DumDBSCL is both negative and highly significant. This confirms our initial hypothesis that provinces in the Mekong Delta (such as An Giang, Dong Thap, and Vinh Long) face structural challenges that naturally drive people away. Beyond standard economic variables, this reflects a powerful "push" factor unique to the region: the transition from traditional agriculture to more modern sectors. As climate change and saline intrusion impact agricultural yields, a significant portion of the rural workforce is compelled to migrate toward industrial hubs in the Southeast seeking more stable livelihoods.

- Non-Linear Urbanization and the "Saturation" Threshold:

Our decision to use Log(Urbanization) in Model 4 proved to be essential. The results indicate that while urbanization initially attracts migrants, its effect is non-linear. In highly developed centers like Ho Chi Minh City, we observe a "saturation point." As urbanization reaches its peak, the attraction of the city is countered by negative externalities such as surging housing prices, traffic congestion, and a higher cost of living. This explains why the "pull" of a city doesn't grow forever—at a certain stage, the marginal benefit of moving to a hyper-urbanized area starts to diminish for many migrants.

- The Persistence of Out-Migration Trends:

The negative and significant coefficient of Omlag (Out-migration Lag) reveals a worrying "momentum" in the southern migration landscape. Instead of a self-correcting mechanism, our findings indicate that out-migration breeds more out-migration. When a province (especially in the Mekong Delta) experiences a high level of out-migration in the previous period, it negatively impacts the current Net Migration. This suggests a "brain drain" or "labor flight" effect: once a significant portion of the young or skilled population begins to leave, it signals a decline in local opportunities, which in turn discourages new people from moving in and prompts remaining residents to follow suit. In provinces like An Giang or Dong Thap, this creates a persistent downward cycle that is difficult to break without major policy intervention.

- Density as the Primary Magnetic Force:

Finally, the robust positive impact of Population Density across all models underscores that "concentration creates opportunity." In the context of Southern Vietnam, density is a proxy for market vibrancy. Even in the face of regional challenges, areas that maintain higher density continue to act as magnets for human capital, suggesting that urban planning policies focusing on high-density, multi-functional zones will be the most effective in managing and attracting migration flows in the future.

7. CONCLUSION & LIMITATION

7.1. Conclusion

In summary, this research successfully identifies the primary factors that drive Net Migration in key Southern Vietnamese provinces. Through a detailed model selection process, Model [4] proved to be the best choice, balancing its ability to explain the data with statistical reliability. Our findings

confirm that migration in this region is a mix between the "pull" of Population Density and the structural "push" of geographical factors, especially in the Mekong Delta.

The study highlights that while urbanization is a key factor, its impact is not linear. This suggests that urban overcrowding is a real challenge for major hubs like Ho Chi Minh City, where the benefits of city life may be canceled out by high costs and traffic congestion. By fixing common technical issues like multicollinearity, this report provides a solid foundation for understanding how population and regional features shape the movement of people in the South. Ultimately, our results suggest that to manage migration effectively, policymakers should consider the unique disadvantages of provinces like An Giang or Dong Thap.

7.2. Limitation

Despite the clear results, our group recognizes several limitations in this study:

- **Scope of Variables:** A primary theoretical driver of migration is income differential (GRDP per capita and wage differences). However, due to data gaps in official GSO reporting for certain Mekong Delta provinces, these were excluded to maintain data integrity. Instead, we utilized Population Density and Urbanization as effective proxy variables. Additionally, our model does not yet account for qualitative factors such as local government policies or environmental issues like saltwater intrusion, which are significant in the Delta region.
- **Potential Endogeneity:** Due to the inclusion of the lagged variable (Omlag), the model may be subject to a degree of endogeneity. While OLS provides a clear baseline for understanding migration momentum within the scope of this undergraduate research, future studies could employ more advanced dynamic panel data techniques to obtain more consistent estimates
- **Sample and Timeframe:** Because of limited data, the research only looks at a specific period. A larger study covering more years would be better to see the long-term trends of migration and the lasting impact of past events (Omlag).
- **Technical Constraints:** While Model 4 passed the primary tests, as sophomore students, we know that more advanced methods, could help us understand even better how migration in one province affects the provinces next to it.

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9. APPENDIX

9.1. Appendix A: R-Studio Regression Outputs

Call:

```
lm(formula = NM ~ log(Urb) + Pd + Omlag + DumDBSCL, data = V8)
```

Residuals:

Min	1Q	Median	3Q	Max
-16.5889	-2.3376	0.0318	2.8816	11.4151

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.0526	10.0330	1.401	0.16544
log(Urb)	-4.1847	2.4843	-1.684	0.09626 .
Pd	3.5654	0.6474	5.508	4.88e-07 ***
Omlag	-0.2519	0.1102	-2.286	0.02509 *
DumDBSCL	-7.7320	2.8410	-2.722	0.00807 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.435 on 75 degrees of freedom

Multiple R-squared: 0.7011, Adjusted R-squared: 0.6851

F-statistic: 43.97 on 4 and 75 DF, p-value: < 2.2e-16

RMSE: 4.294594

MAE: 3.211971

MAPE: 75.04908

9.2. Appendix B: Post-Estimation Diagnostic Plots

- RESET test

data: find6

RESET = 2.3652, df1 = 2, df2 = 73, p-value = 0.1011

- Breusch-Pagan test

data: find6

BP = 7.2615, df = 4, p-value = 0.1227

9.3. Appendix C: Data Sample Details

Province	Year	Net Migration (%)	Population density (thousands people/km2)	Unemployment rate (%)	Urban ratio (%)	Interaction (Unemployment * Urban)	Out_migration_lag(%)	Interaction (Population density * Urban)
An Giang	2015	-9,7	0,566	2,33	30,32	70,6456	0	17,16112
	2016	-9,1	0,559	1,88	30,65	57,622	11,8	17,13335
	2017	-3,7	0,553	2,24	30,99	69,4176	10,4	17,13747
	2018	-9,8	0,546	2,51	31,33	78,6383	5	17,10618
	2019	-14,4	0,539	3,18	31,59	100,4562	11,9	17,02701
	2020	-7,1	0,556	2,76	31,58	87,1608	15,6	17,55848
	2021	-2,4	0,558	2,88	33,83	97,4304	12,5	18,87714
	2022	-5,1	0,56	2,23	33,84	75,4632	6,6	18,9504
	2023	-6,1	0,56	2,45	37	90,65	9,5	20,72
	2024	-3,7	0,561	2,24	36,94	82,7456	12	20,72334